Sample Abstract Format

Title: ANALYSIS OF MAGNETRON DEPOSITION OF SANDWICH MULTILAYER COATING ON CHROME-NICKEL STEEL

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ABSTRACT:

A quality of a tool parts coating can ensure a quality workpiece surface for both users and manufacturers. The idea of combining coatings into a multilayer was developed in response to industry demand to improve the mechanical properties of the resulting coating. The substrate material chosen was 100Cr6 chromium bearing steel. The aim is to analyze the results of measurement methods of magnetron deposition of surface TiAlN and TiB2 multilayers and their evaluation. The issues of PVD deposition, magnetron sputtering, specifically the HiPIMS method, which is the method of deposition of the investigated multilayer. are described.
The aim of the research activity was to prove the improvement of the properties of the multilayer while maintaining the conditions. Firstly, the individual TiAlN and TiB2 coatings and their composition were analyzed using glow discharge spectroscopy and X-ray diffraction. Then the multilayer was analyzed. There were positive findings, such as a really significant increase in the microhardness of the coated layers. The microhardness was double that of the reference uncoated sample. In addition, the thickness of the individual coatings and the resulting multilayer conforms to the requirements and thus there is no influence on the dimensional properties. The linear analysis showed a good interconnection between the individual coatings as well as between the coating and the substrate. Tribological testing revealed an improvement in abrasion resistance. It can also be concluded from this test that the improvement in the mechanical properties mentioned above occurred while maintaining the frictional conditions. Thus, the deposition of these coatings on top of each other in a sandwich multilayer is possible, there is an excellent interlocking of the coatings for improved mechanical properties. The choice of TiAlN provides toughness, while the ceramics - TiB2 coating provides hardness.

BIOGRAPHY:

He works as an associate professor at the Faculty of Mechanical Engineering of the Jan Evangelista Purkyne University in Ústí nad Labem, where, in addition to teaching Physics of Metals and Technical Physics and working with students, he is engaged in research and publishing activities. He has written several publications and university textbooks. He is the author of professional papers in scientific and professional journals and proceedings at home

and abroad, and he also teaches classes for foreign students at the University. Also is author of several technical patents ativly applied into technical pactice.
His research activities are focused on the analysis and preparation of micro- and nano-sized particles suitable for coating metal alloys and mechanical alloying.

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